MIL-F-43974
23 February 1978
See 6.4

MILITARY SPECIFICATION

FLAVORING, LEMON CRYSTALS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirement for Flavoring, Lemon Crystals, for use by the Armed Forces as an item of general issue.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit: Preparation of Nonperishable Subsistence

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, ATTN: DRXNM-TDS, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8950

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:15Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

(Copies of specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

US DEPARTMENT OF AGRICULTURE

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Technical Division, Food Safety and Quality Service, US Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

THE UNITED STATES PHARMACOPOEIAL CONVENTION, INC.

Pharmacopoeia of the United States

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:15Z
Check the source to verify that this is the current version before use.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, DC 20044.)

NATIONAL RESEARCH COUNCIL

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

E380 Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(*Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.*)

3. REQUIREMENTS

3.1 Bid sample approval. Duplicate, individually packaged samples, consisting of not less than 8 ounces (227 g) of lemon crystals, which the bidder proposes to furnish and which comply with the specification requirements, shall be submitted to the contracting officer prior to the bid opening and will be inspected to the extent necessary to properly evaluate the bids. One sample of each duplicate shall be evaluated to determine compliance with 3.3 for the subjective characteristics (flavor, odor, color). In addition, the sample shall be rated as to overall quality, taking into consideration all the subjective characteristics. The remaining sample shall be used as a standard for the subjective characteristics in determining acceptance of intended delivery products. The approval of any bid sample for the aforementioned characteristics shall not constitute approval of the product as meeting the other requirements of this specification.

3.2 Material. All materials shall be clean, sound, wholesome and free from any foreign odors, flavors or extraneous matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:15Z
Check the source to verify that this is the current version before use.

3.2.1 Oil of lemon. Oil of lemon shall meet the requirements of the Food Chemicals Codex and shall be cold-pressed lemon oil of single strength or partially deterpenized oil prepared therefrom or a combination of these. They shall be entrapped in a suitable carbohydrate carrier.

3.2.2 Dehydrated lemon juice. Dehydrated lemon juice shall be produced from natural lemon juice or frozen concentrate. It may contain processing aids such as corn syrup solids, emulsifiers or sulfur dioxide, which shall meet the requirements of Food Chemicals Codex, if referenced therein.

3.2.3 Acids. Food acids shall conform to the requirements of the Food Chemicals Codex.

3.2.4 Buffers. Buffers shall include only those of Food Chemicals Codex quality, such as potassium citrate.

3.2.5 Tricalcium phosphate. Tricalcium phosphate shall be of anhydrous grade of Food Chemicals Codex quality, when used as an anticaking agent.

3.2.6 Lactose. Lactose shall contain 99 percent lactose. It shall comply with the requirements of United States Pharmacopoeia.

3.2.7 Coloring. Coloring materials shall be vegetable colors or approved US certified food colors and shall be acid resistant.

3.2.8 Thickeners or cloud agents. Thickeners or cloud agents such as gum arabic (acacia) shall be used.

3.2.9 Antioxidants. Antioxidants shall comply with Food Chemicals Codex.

3.3 Finished product. The finished product shall be uniformly blended, free from foreign materials, undesirable flavors and odors, burnt or scorched particles and lumps which will not break under light finger pressure. The product (2 oz) (56 g) shall dissolve in 32 fluid ounces (0.95 liters) of cold tap water after 1 minute of stirring to produce a light yellow to yellow colored beverage containing a slight cloudiness (turbidity) similar to fresh lemon juice. The reconstituted product shall possess a lemon-like flavor.

3.3.1 Mandatory ingredients. Mandatory ingredients are: dehydrated lemon juice, lemon oil, antioxidants such as BHA, acid, cloud agents and coloring.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:15Z
Check the source to verify that this is the current version before use.

3.3.1.1 Optional ingredients. Optional ingredients may be carbohydrates and buffering agents, as specified.

3.3.2 Analytical requirements. The finished product shall meet the requirements of table I.

TABLE I. Analytical requirements

Requirement	Percent or as specified
Moisture	3 percent (maximum)
Volatile oils	ml/100 g minimum
Titratable acidity (dry powder)	35-42 percent (as anhydrous citric acid)
Reducing sugar	40 percent maximum (expressed as lactose)
pH (10 percent w/v)	pH value 2.5 - 2.7

3.4 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder.

3.5 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be prepared in plants meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection. Product shall be rejected if produced in plants not meeting the requirements of 3.5.

4.3 Inspection.

4.3.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the US Department of Agriculture, the United States Standards for Condition of Food Containers shall apply.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:15Z
Check the source to verify that this is the current version before use.

4.3.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.3 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3.3.1 Examination of ingredients. Conformance of ingredients to identity requirements (kind, quality standards, composition, as applicable) shall be ascertained by examination of pertinent labels, invoices or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to condition requirements of 3.1 and as may be contained in the applicable ingredient requirement paragraph. The sample unit for examination shall be 8 ounces (227 g) of ingredient from a primary container. The sample size shall be five. In the event a lot consists of less than five containers, a sample shall be extracted from each container in the lot. Nonconformance with identity requirements or one or more sample units not conforming to condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

4.3.3.2 Testing of cans. The cans shall be tested for tin plate to determine conformance to 5.1.1. Test procedures shall be in accordance with 4.4. The sample unit shall be one can with lid and the sample size shall be the number of sample units indicated by inspection level S-1. Lot size shall be expressed in cans. Test requirements shall be an average requirement. The test results shall be reported to the nearest 0.01 pound (0.1 g) tin plate. A test failure shall indicate unacceptable cans and use of such shall be cause for rejection of the involved end item.

4.3.4 Examination of end item. Examination of end item shall be in accordance with tables III and IV and 4.3.4.1 through 4.3.4.3. The sampling for the aforementioned tables shall be in accordance with table II. The acceptable quality levels (AQLs) shall be expressed as percent defective for table III and defects per hundred units for table IV.

TABLE II. Sampling plans

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major	AQLs Minor
III	One filled can	cans	S-3	-	2.5
IV	One filled can	cans	S-2	1.5	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:15Z
Check the source to verify that this is the current version before use.

TABLE III. Examination of net weight 1/

Category	Defect
Minor	
201	Less than 7-13/16 ounces (222 g)

1/ Weighing device shall be calibrated in increments no larger than 1/16 ounce (2 g). Lot shall be rejected if sample data indicate a lot average net weight of less than 8 ounces (227 g).

TABLE IV. Examination of product 1/

Category	Defect
Major	Finished product
101	Not uniformly blended
102	Lumps that do not fall apart with light finger pressure
103	Presence of scorched or burnt particles
104	Presence of undesirable flavors or odors
	Prepared product 2/
105	Does not go completely into solution within the prescribed time limit
106	Color not similar to fresh lemon juice
107	Not equal to or better than bid sample, when applicable

1/ Presence of foreign material (e.g., metal, wood, glass, paint, filth, insects, insect parts, etc.) shall be cause for rejection of the lot.

2/ Preparation of beverage for sample unit and bid sample shall be in accordance with 3.3. Make one determination per sample unit.

4.3.4.1 Examination of cans. Examination of cans for external condition and labeling shall be in accordance with the United States Standards for Condition of Food Containers. In addition, the label shall be examined to determine conformance to 3.3.1. One or more mandatory ingredients not listed on the label shall be cause for rejection of the lot.

4.3.4.2 Examination for packing and marking. Examination shall be in accordance with the quality assurance provisions of PPP-C-29.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:15Z
Check the source to verify that this is the current version before use.

4.3.4.3 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078, and the applicable specification sheet for the type and class of load specified.

4.3.5 Testing of finished product. The finished product shall be tested to determine conformance to table I. Test procedures shall be as specified in 4.4. The sample for testing shall be an 8 ounce (227 g) composite of the product derived from the number of cans indicated by inspection level S-2. Lot size shall be expressed in terms of cans. Test results shall be reported to the same unit or decimal place as specified for the pertinent requirements. A test failure shall be cause for rejection of the lot.

4.4 Examination and test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the contractor if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor will use the procedures set forth herein. In case of dispute as to examination of test results, the procedures specified herein will govern.

4.4.1 Examination procedure.

4.4.1.1 Can leakage. Can leakage shall be determined in accordance with the procedures in PPP-C-29.

4.4.2 Test procedures.

4.4.2.1 Chemical analyses. Chemical analyses shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists:

Test	Source		Method
Moisture	Chapter: Section:	Sugars and Sugar Products Sugars and Sirups	Drying upon quartz sand
Volatile oils	Chapter: Section:	Fruits and Fruit Products Essential Oil	Essential oil
Titratable acidity	Chapter: Section:	Fruits and Fruit Products Acidity	Titratable acidity
Reducing sugar	Chapter: Section:	Fruits and Fruit Products Sucrose	Reducing sugars before and after inversion
pH	Chapter: Section:	Beverages: Wines Acids	pH

4.4.2.2 Tin coating weight. Tin coating weight shall be determined by any method specified in PPP-C-96.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:15Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation-packaging. The product shall be packaged in accordance with level A or C, as specified (see 6.1).

5.1.1 Level A. A net quantity of 8 ounces (227 g) of product shall be filled into a size 211 by 400 open top style, round, metal can with soldered side seam and compound-lined, double-seamed ends. A minus 3/16 ounce (5 g) tolerance will be allowed in any individual container provided the average net quantity of the container, examined in accordance with table III, is not less than 8 ounces (227 g). The can shall be made throughout from not less than commercial 0.25 pound (113 g) electrolytic tin plate per base box (see 4.4.2.2) and shall be coated overall on the outside with a coating conforming to type I of TT-C-495. The top of can shall be suitably scored to provide a pull type, full panel opening. A corrosion resistant finger grip shall be affixed to one edge of the top panel. The can shall be provided with a plastic lid for reclosure after partial use of the contents. The filled and sealed can shall not leak when examined in accordance with 4.4.1.1.

5.1.2 Level C. The product shall be packaged in accordance with 5.1.1, except that cans with or without commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C requirements of PPP-C-29 for size 211 by 400 cans, as specified (see 6.1).

5.3 Unitized loads. When specified (see 6.1), the shipping container of the product shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Marking.

5.4.1 Cans. Cans shall be labeled in accordance with MIL-L-1497. Labeling shall contain the following information:

DIRECTIONS FOR USE

1. To make 1 gallon lemon-flavored beverage (unsweetened)

   Dissolve 1 can (8 ounces) lemon crystals into 1 gallon water.

2. To replace fresh lemon juice

   Substitute lemon beverage ounce for ounce to replace fresh lemon juice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:15Z
Check the source to verify that this is the current version before use.

DIRECTIONS FOR USE (cont'd)

3. To make lemon filling for pies and pastries.

Substitute lemon beverage ounce for ounce to replace the fresh lemon juice in the recipe.

4. To make lemonade.

Make 4 gallons lemon beverage and dissolve 7 pounds sugar in it.
Refrigerate if not used immediately.
Before serving add 3 gallons of chopped ice or ice cubes (12 pounds)

YIELD: 100 Servings

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Unitized loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Levels of packaging and packing required (see 5.1 and 5.2).
(c) Type and class of unit load when unit loading is specified (see 5.3).

6.2 Intended use. Lemon crystals are currently used as a flavoring agent, substitute for lemon juice and for bakery products such as lemon pie filling.

6.3 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/ AFR 71-6/MC 4030.14D/DSAR 4145.7.

Source: https://assist.dla.mil -- Downloaded: 202640-21T23:15Z
Check the source to verify that this is the current version before use.

6.4 Supersession data. This document includes the requirements in IP/DES S-1-4, dated 17 April 1974.

6.5 Metric equivalents. Metric equivalents, indicated in parenthesis throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8950-0255

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

Flavoring, Lemon Crystals MIL-F-43974

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:15Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK RESEARCH and DEVELOPMENT COMMAND
NATICK, MASSACHUSETTS 01760

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
DRXNM-TD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD-314

U.S.MAIL

Commander
US Army Natick Research and Development Command
ATTN: DRXNM-TD
Natick, MA 01760

NATICK OP 1 JUL 77

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:15Z
Check the source to verify that this is the current version before use.